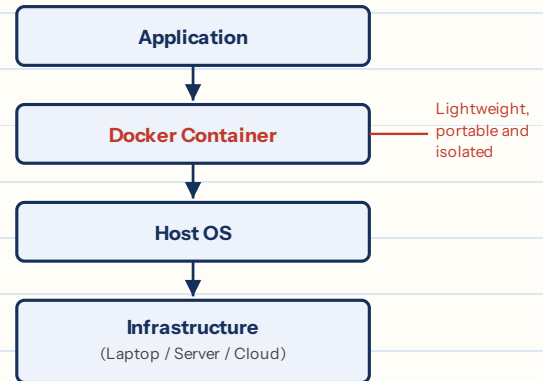


DOCKER NOTES

– Page 1 –

1. What is Docker?

Docker is an open-source platform that helps us to build, ship and run applications inside containers. It allows an application to run consistently on any environment.



2. Why Docker was created?

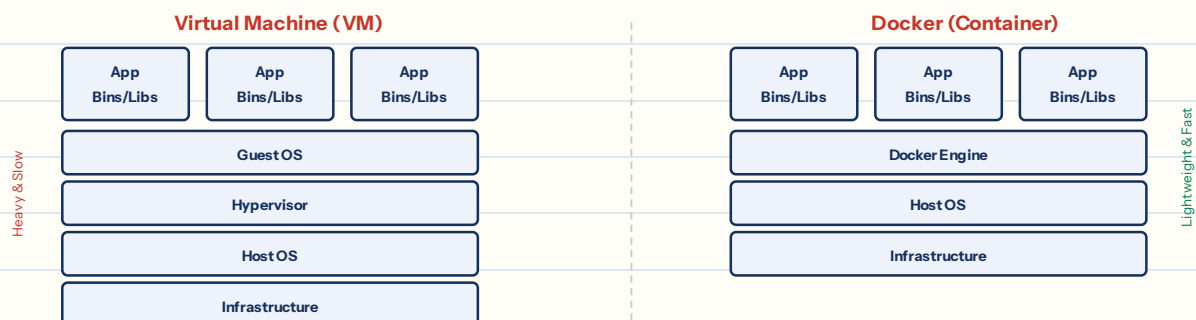
Before Docker, developers faced many issues while deploying applications:

- ★ "It works on my machine" problem
- ★ Heavy virtual machines
- ★ Slow deployment
- ★ Different environments
- ★ Complex configuration

3. Benefits of Docker

- ★ **Lightweight** — Containers are smaller and use fewer resources.
- ★ **Portable** — Build once and run anywhere.
- ★ **Consistent** — Same environment everywhere.
- ★ **Fast Deployment** — Start containers in seconds.
- ★ **Isolation** — Each container runs isolated from others.
- ★ **Scalability** — Easy to scale applications.

4. VM vs Docker

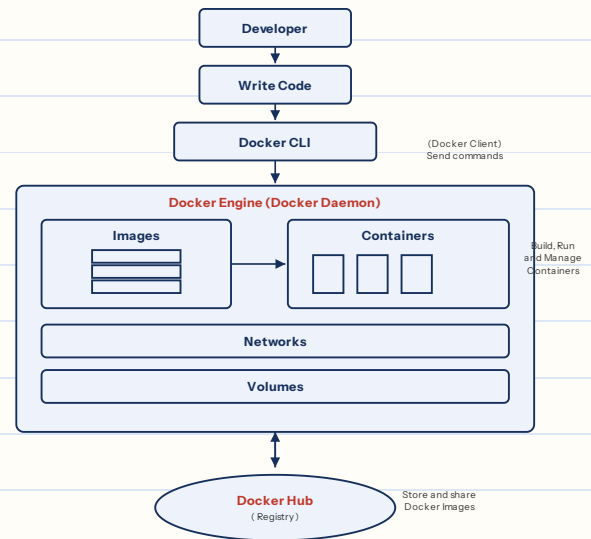


1. Docker Architecture

Docker follows a client-server architecture. Docker Client communicates with Docker Daemon (Docker Engine) which builds, runs and manages Docker containers.

Components

- **1. Docker Client (CLI)** — command line tool used to send commands to Docker Daemon.
- **2. Docker Daemon (Engine)** — core part of Docker; builds images, runs containers, manages networks & volumes.
- **3. Docker Images** — read-only templates used to create containers.
- **4. Docker Containers** — running instances of images; lightweight, portable, isolated.
- **5. Docker Hub (Registry)** — cloud based registry service; push/pull images.
- **6. Networks** — communication between containers & outside world. Types: bridge, host, none, overlay.
- **7. Volumes** — store data generated by containers; persists even after removal.

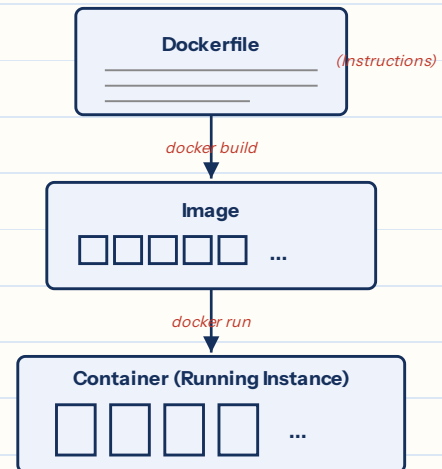


How it works?

We give commands through Docker CLI → It talks to Docker Daemon → Daemon pulls images from Docker Hub (if not present) → Creates and runs containers → Manages Networks and Volumes.

1. Image vs Container vs Dockerfile

Docker uses images to create containers. Image is a blueprint (read-only template) and container is a running instance of it. Dockerfile is a text file which has instructions to build an image.



2. Image vs Container

Feature	Image	Container
Definition	Blueprint or template (read-only)	Running instance of image
State	Static	Dynamic
Change	Cannot be changed	Can be changed
Life	Permanent until deleted	Temporary — can be stopped/removed
Size	Larger	Smaller
Purpose	To create containers	To run applications
Examples	ubuntu:22.04, mysql:8, nginx:latest	Running nginx, mysql container

3. Docker Lifecycle

A simple flow of how Dockerfile, Image and Container work together.



4. Key Points

- ★ Image is built once and can be used to create multiple containers.
- ★ Containers are isolated, portable and lightweight.
- ★ Docker ensures "Build once, Run anywhere".
- ★ Containers share the host OS kernel, that's why they are lightweight.

1. Dockerfile Explained

A Dockerfile is a text file that contains a set of instructions to build a Docker image automatically.



2. Sample Dockerfile

```

FROM openjdk:17-jdk-slim # Base image

WORKDIR /app # Set working directory

COPY app.jar . # Copy file to container

ENV APP_ENV=production # Set environment variable

EXPOSE 8080 # Expose port

CMD ["java", "-jar", "app.jar"] # Command to run app
  
```

Note

Each instruction creates a layer in the image. If a layer doesn't change, Docker uses cache to build faster.

3. Important Instructions

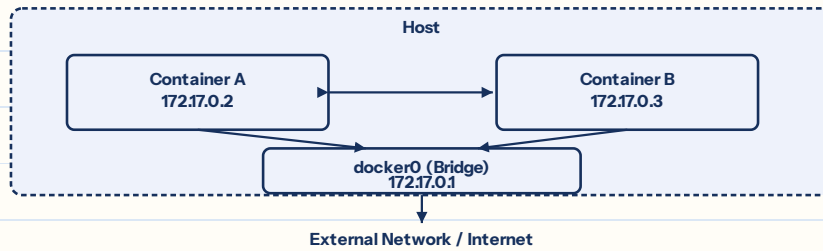
- **FROM** — Specifies the base image to start with.
- **WORKDIR** — Sets the working directory inside the container.
- **COPY** — Copies files or directories from host to container.
- **ADD** — Similar to COPY, also supports URLs and auto-extracts .tar files.
- **ENV** — Sets environment variables inside the container.
- **EXPOSE** — Informs Docker that the container listens on the given port.
- **CMD** — Specifies the default command to run when container starts.
- **ENTRYPOINT** — Configures a container that will run as an executable. It allows the container to run as if it was that executable.

4. CMD vs ENTRYPOINT

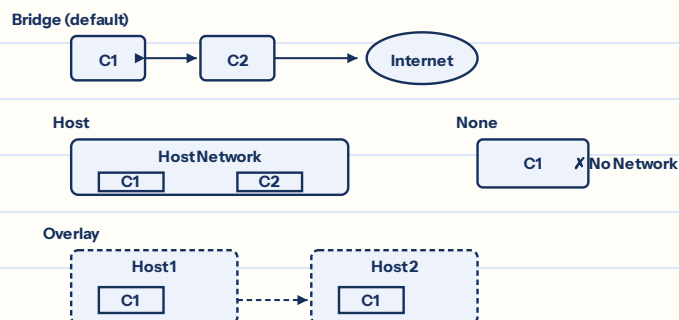
CMD	ENTRYPOINT
Provides default command and arguments.	Configures the container to run as executable.
Can be overridden when running container.	Cannot be overridden easily.
Only one CMD instruction is respected.	Only one ENTRYPOINT instruction is respected.
Used mostly for defining default behavior.	Used when the container must always run a specific program.

1. Docker Networking

Docker containers can communicate with each other and with outside world using networks.



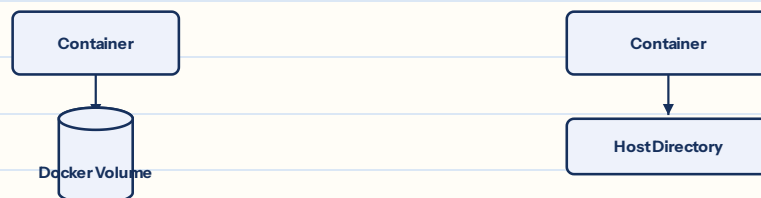
Types of Networks



- ★ **Bridge (default)** — containers on the same bridge network can talk; can access outside network.
- ★ **None** — container has no network access. Useful for complete isolation.
- ★ **Host** — container shares the host network. No isolation at network level.
- ★ **Overlay** — connect containers across multiple Docker hosts (Swarm mode).

2. Storage in Docker

Containers are ephemeral. When a container is removed, its data is lost. To persist data we use Volumes or Bind Mounts.

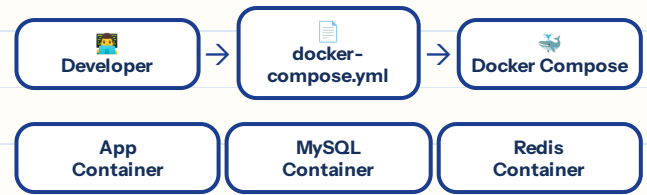


Feature	Volume	Bind Mount
Managed by	Docker	Host
Location	Inside Docker host (/var/lib/docker/volumes)	Anywhere on host system
Portability	High — platform independent	Lower (depends on host path)
Backup	Easy (docker volume backup)	Manual
Best for	Production	Development (code changes reflect immediately)

1. Docker Compose

Docker Compose is a tool for defining and running multi-container Docker applications. With a single YAML file, we can configure all the services, networks and volumes required for our app. It makes it easy to start, stop and manage the entire application.

How it works?



2. docker-compose.yml (Example)

```

version: '3.8' # Compose file version
services: # All services
  app: # Service name
    image: myapp:latest # Image to use
    ports: # Port mapping host:container
      - "8080:8080"
    depends_on: # Start mysql before app
      - mysql
  mysql: # Another service
    image: mysql:8 # MySQL image
    environment: # Environment variable
      MYSQL_ROOT_PASSWORD: root
    volumes: # Persist data in volume
      - db_data:/var/lib/mysql
  redis: # Another service
    image: redis:latest # Redis image
volumes:
  db_data: # Named volume
  
```

3. Important Commands

- ★ **docker compose up** → Build and start all services defined in the file.
- ★ **docker compose up -d** → Start in detached mode (background).
- ★ **docker compose down** → Stop and remove all containers, networks.
- ★ **docker compose logs** → View logs of all services.
- ★ **docker compose ps** → List all running services.
- ★ **docker compose build** → Build or rebuild services.

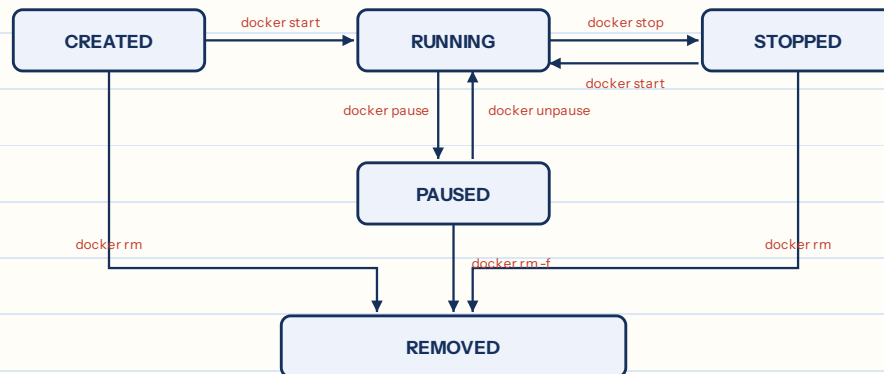
4. Multi-container Application Example



Default Network (created by Compose)

1. Docker Container Lifecycle

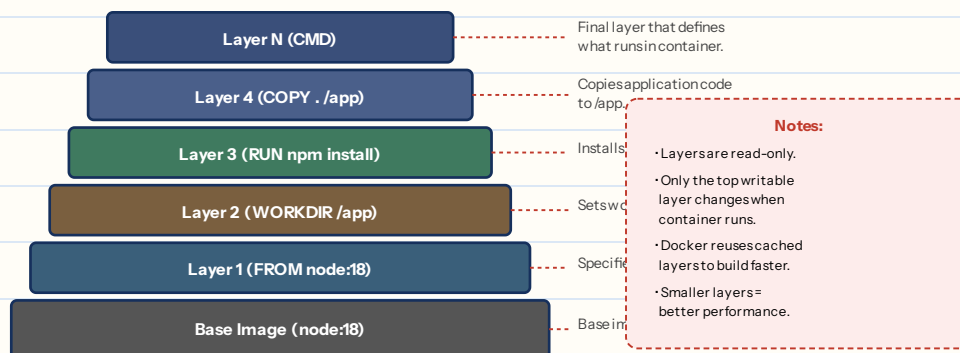
A container goes through different states during its lifetime.



- ★ **Created** — Container is created but not started.
- ★ **Paused** — Container is running but all processes are paused.
- ★ **Removed** — Container is deleted and cannot be recovered.
- ★ **Running** — Container is up and running.
- ★ **Stopped** — Container is stopped but still exists.

2. Docker Image Layers

Docker images are built using a layered architecture. Each instruction in the Dockerfile adds a new layer.



How Layer Caching Works?

First Build	Code change in Step 3	Rebuild
Step 1 ✓	Step 1 ✓	Step 1 (cached)
Step 2 ✓	Step 2 ✓	Step 2 (cached)
Step 3 ✓	Step 3 (changed)	Step 3 (rebuilt)
Step 4 ✓	Step 4	Step 4 (rebuilt)

Only the changed layer and the layers above it are rebuilt. Others are reused from cache.

- Page 8 - Docker Commands Cheat Sheet -

Here are the most commonly used Docker commands every developer / tester should know.

Command	Description	Example
<code>docker --version</code>	Check Docker version	<code>docker --version</code>
<code>docker info</code>	Display system-wide information	<code>docker info</code>
<code>docker login</code>	Login to Docker registry	<code>docker login</code>
<code>docker logout</code>	Logout from Docker registry	<code>docker logout</code>
<code>docker search <image></code>	Search image on Docker Hub	<code>docker search nginx</code>
<code>docker pull <image></code>	Download image from registry	<code>docker pull nginx:latest</code>
<code>docker images</code>	List all downloaded images	<code>docker images</code>
<code>docker rmi <image-id></code>	Remove an image	<code>docker rmi nginx</code>
<code>docker build -t <name> .</code>	Build image from Dockerfile	<code>docker build -t myapp:1.0 .</code>
<code>docker run <image></code>	Run a container	<code>docker run nginx</code>
<code>docker run -d <image></code>	Run container in detached mode	<code>docker run -d nginx</code>
<code>docker ps</code>	List running containers	<code>docker ps</code>
<code>docker ps -a</code>	List all containers (running + stopped)	<code>docker ps -a</code>
<code>docker start <container-id></code>	Start a stopped container	<code>docker start alb2c3d4</code>
<code>docker stop <container-id></code>	Stop a running container	<code>docker stop alb2c3d4</code>
<code>docker restart <container-id></code>	Restart a container	<code>docker restart alb2c3d4</code>
<code>docker rm <container-id></code>	Remove a stopped container	<code>docker rm alb2c3d4</code>
<code>docker rm -f <container-id></code>	Force remove a running container	<code>docker rm -f alb2c3d4</code>
<code>docker logs <container-id></code>	View logs of a container	<code>docker logs alb2c3d4</code>
<code>docker exec -it <container-id> bash</code>	Access container shell	<code>docker exec -it alb2c3d4 bash</code>
<code>docker inspect <container-id></code>	Inspect container details	<code>docker inspect alb2c3d4</code>
<code>docker cp <src> <container>:/dest</code>	Copy files/folders to container	<code>docker cp app.py alb2c3d4:/app</code>
<code>docker cp <container>:/src <dest></code>	Copy files/folders from container	<code>docker cp alb2c3d4:/app /home/user</code>
<code>docker network ls</code>	List all networks	<code>docker network ls</code>
<code>docker volume ls</code>	List all volumes	<code>docker volume ls</code>
<code>docker system prune</code>	Remove unused data (images, containers)	<code>docker system prune</code>
<code>docker stats</code>	Show live resource usage of containers	<code>docker stats</code>
<code>docker tag <image> <new-tag></code>	Tag an image	<code>docker tag myapp:1.0 myapp:latest</code>
<code>docker push <image></code>	Push image to registry	<code>docker push myapp:1.0</code>